EcoHive Kenya Ltd. — Climate-Smart Honey Value Chain

Production URL: https://ecohive-jet.vercel.app/
Development & Live Mirror: https://ais-pre-vegkej4u2lmpqczdp2pstl-122122115705.europe-west2.run.app
Capstone Track: Frontend AI Engineering & General AI Fluency (Week 8)
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Assignment Code: FE-11 & FL-09

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1. What It Does and For Whom

EcoHive Kenya Ltd. is a climate-tech social enterprise modernizing East African apiculture.

The Problem

Traditional log and timber beekeeping in Kenya results in massive deforestation, low honey yields (<8kg/hive/year), high colony absconding rates (>40%) due to erratic climate shocks, and exploitative middleman pricing that leaves rural smallholders in poverty.

The Solution

EcoHive introduces a circular, technology-enabled honey value chain:

1. **Recycled Composite Beehives**: 100% recycled UV-stabilized HDPE plastic blended with high-insulation agricultural fiber waste. 25-year lifespan, impervious to pests, termite-proof.
2. **Solar GSM IoT Telemetry**: Solar-powered sensor nodes measuring hive temperature, total colony weight (for harvest timing), and acoustic frequency (for swarming detection).
3. **Traceable Fair-Trade Value Chain**: Direct market linkage for over 1,200 smallholder beekeepers with transparent export-grade quality certification.

For Whom

- **Rural Smallholder Farmers & Cooperatives**: Real-time harvest alerts, subsidized durable hives, and guaranteed off-take contracts.
- **Impact Investors & ESG Funds**: Verifiable carbon and plastic diversion metrics (15+ tons plastic recycled), rural job creation, and export revenue models.
- **Bulk Honey Importers & Retailers**: Batch-traceable raw organic acacia honey, medical-grade propolis, and beeswax.

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2. Interactive Systems Overview

Module	Purpose & User Experience	Tech Stack
Signature Shader Hero	Interactive WebGL fluid caustics simulating Kenyan honey ripples responding to cursor vectors.	Raw WebGL, GLSL, Simplex Noise
Interactive 3D Smart Hive	Procedural 3D Langstroth hive with box explosion animations, material configurator, and raycasted parts inspection.	Three.js, React 19, TypeScript
Live IoT Apiary Telemetry	Real-time monitoring of Baringo & Nakuru apiaries (temperature, weight, acoustic status).	Express REST API, React Hooks
Hive AI Assistant	Multi-turn conversational agent with 4 role personas (Farmer, Investor, Buyer, Guide), dynamic follow-up chips, and dual speed/depth model switching.	Google Gemini 3.5 Flash & 3.1 Flash-Lite, Node.js
Telemetry & Privacy Analytics	Zero-cookie client-side telemetry dashboard tracking HTTPS health and pageview trends.	Custom Express middleware

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3. Architecture Sketch

```
                      [ Client Browser / Mobile ]
                                   │
             ┌─────────────────────┴─────────────────────┐
             ▼                                           ▼
    [ React 19 + Tailwind ]                     [ Three.js Canvas ]
  • Signature GLSL Shader Hero                • Procedural 3D Hive Model
  • Multi-Tab View Router                     • Raycasting Part Inspector
  • Multi-Turn Hive AI Drawer                 • Box Explosion Controller
  • Privacy Analytics Tracker
             │
             ▼ HTTP / JSON REST
   [ Node.js + Express Server ] (Port 3000)
  ├── Rate Limiting & Input Caps Guard (Sliding Window per IP)
  ├── /api/iot-telemetry ──────> Real-time Sensor Interpolation
  ├── /api/contact ────────────> Lead Routing & Validation Engine
  ├── /api/analytics/track ────> Privacy Telemetry Event Store
  └── /api/ai-assistant ───────> Multi-Turn Gemini Engine (3.5 Flash / 3.1 Flash-Lite)
```

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4. Environment Variables

Variable	Required?	Purpose	Default / Fallback
`PORT`	Optional	Server listening port	`3000`
`NODE_ENV`	Optional	Runtime environment mode	`development`

Variable	Required?	Purpose	Default / Fallback
`GEMINI_API_KEY`	Optional	Powers the Hive AI assistant	If missing, system gracefully falls back to deterministic corporate routing

(Note: Create a `.env` file in the root directory following `.env.example`).

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5. Step-by-Step Setup Guide (Clone & Run in 60s)

Prerequisites

- **Node.js**: v18.0.0 or higher
- **npm**: v9.0.0 or higher

1. Clone the repository

```
git clone https://github.com/munyieric7/ecohive-kenya.git
cd ecohive-kenya
```

2. Install dependencies

```
npm install
```

3. Configure environment variables (optional)

```
cp .env.example .env
# Add your GEMINI_API_KEY if testing live AI inference
```

4. Start development server

```
npm run dev
```

The application will boot on `http://localhost:3000` with hot-module reloading and full API mocking.

5. Production build & bundle

```
npm run build
npm start
```

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6. Engineering Decisions

1. **Procedural 3D Geometry over Heavy GLB Meshes**:
   - *Decision:* Instead of downloading a 15–20MB 3D model asset over Kenyan mobile networks, the beehive anatomy is procedurally constructed using raw Three.js primitives.
   - *Outcome:* Zero network payload for 3D assets, <35ms parse time, and instant 60fps rendering on mobile.
2. **Dual-Tone Visual System**:
   - *Decision:* Warm honey gold (`#F59E0B`) paired with off-white (`#FDFBF7`) and deep slate text (`#1C1917`).
   - *Outcome:* Balances agricultural warmth for rural farmers with clean typography for international ESG investors.
3. **Privacy-First In-App Telemetry**:
   - *Decision:* Rather than loading invasive 3rd-party trackers (e.g. Google Analytics or Meta Pixel), we built a lightweight in-memory event store with an in-app viewer.
   - *Outcome:* 0ms cookie consent friction, GDPR/Kenyan Data Protection Act compliance, and instant proof of launch.

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7. Production Hygiene & Abuse Protection (FE-11)

To protect the server and API tokens against abuse, loops, or credit depletion:

- **IP Sliding-Window Rate Limiting**: Production sliding window allowing up to **30 requests per minute** per client IP on `/api/ai-assistant`, supporting rapid multi-turn conversation while preventing automated flooding.
- **Strict Input Character Caps**: Messages are strictly clamped to **1,000 characters** max with history truncation to keep token consumption bounded. Payloads exceeding this immediately receive `HTTP 400 Bad Request`.
- **Multi-Tier Cascade & Circuit Breaker**: An intelligent timeout guard and cascading model pipeline (`gemini-3.5-flash` → `gemini-3.1-flash-lite` → `gemini-flash-latest` → domain-grounded offline knowledge base) ensures zero blank screens or crashed threads even during upstream API outages.
- **HTTP 429 Responses**: Returns structured JSON with `Retry-After` header indicating seconds until unlock.

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8. Honest "How AI Tools Built This" (Transparency Diligence)

In accordance with the General AI Fluency Framework (Transparency Diligence):

- **What AI Generated**:
  - Scaffolding the raw GLSL fragment shader simplex noise algorithms and coordinate transforms.
  - Rapid initial structuring of TypeScript data dictionaries (`ecohiveData.ts`).
  - Drafting initial regex patterns for contact form validation.
- **What I Engineered, Checked, and Audited Myself**:
  - **Build Pipeline & CommonJS Fix**: Resolved Vite/esbuild module bundling conflicts to ensure zero-crash production builds on Vercel and Cloud Run.

  - **Contrast & Visual Polish**: Tuned the shader's optical transparency and ambient base colors to enforce strict **>7:1 WCAG AAA** contrast against all typography.
  - **Production Hardening**: Wrote the sliding-window IP rate limiter, character clamping logic, and error handlers.
  - **3D Hive Tuning**: Fine-tuned the Three.js materials, camera clamping angles, and mobile touch event handlers.

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9. Known Limitations

1. **Simulated Telemetry Feed**: The IoT hive telemetry endpoint currently returns simulated sensor readings with natural variance. Physical deployment requires connecting our backend to real MQTT brokers receiving LoRaWAN packets from Baringo apiaries.
2. **In-Memory Store Persistence**: Lead inquiries and analytics events reside in Express in-memory arrays. In enterprise production, these will synchronize to a PostgreSQL / Supabase cluster.

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10. Verification & Audit Results

- **Lighthouse Performance**: 98/100
- **Accessibility**: 100/100 (high-contrast text, focus states, aria-labels)
- **SEO & Social Share**: Complete OpenGraph, Twitter 1200x630 preview cards, and Schema.org `Organization` metadata.
- **FlyRank Graduate Badge**: Verified integration in site footer linking to `https://internship.flyrank.ai/verify?first_name=Eric`.